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Design and Validation of a Real-Time Maintenance Monitoring System Using BIM and Digital Twin Integration

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Abstract: This study presents a real-time monitoring system integrating Building Information Modeling (BIM) and digital twin technology to enhance maintenance efficiency and safety in urban infrastructure. Unlike conventional periodic inspections, which miss dynamic changes and increase costs, this system uses a BIM model at LOD 400 for a solar-powered noise barrier tunnel integrated with the Wansan Tunnel in South Korea. It incorporates IoT sensor data, including vibration, tilt, light, air quality, and water detection, which are synchronized via the Autodesk Forge API, and WebSockets and visualized on a web-based dashboard. A demonstration from 22 October to 7 November 2024 confirmed that this system had stable data transmission, with light sensor rates exceeding 90%, and enabled the detection of anomalies such as irregular illuminance and structural shifts, thereby supporting informed maintenance decisions. While it is proven that BIM–digital twin integration improves NBT management, partial air quality data gaps highlight areas for refinement. This framework lays the groundwork for predictive maintenance through advanced analytics.

Keywords: real-time monitoring; building information modeling (BIM); digital twin; noise barrier tunnel (NBT); maintenance efficiency; urban infrastructure management

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1. Introduction

1.1. Research Background and Context

Smart cities are increasingly recognized as a model capable of addressing future climate challenges driven by rapid urbanization, prompting the development of related research and business models [1]. Concurrently, the Fourth Industrial Revolution has introduced digital twin technology—leveraging IoT, AI, and big data—to drive transformative innovations in construction and infrastructure management [2]. The concept of the digital twin was first proposed by Grieves in a 2002 lecture [3] and was later refined into a more advanced framework by NASA in 2010 [4]. Within smart cities, digital twins are being increasingly applied to fields such as urban planning, traffic management, and environmental monitoring [2]. Notably, ongoing research actively explores the integration of real-time data with Building Information Modeling (BIM) for infrastructure management [5,6]. BIM serves as an effective data format for managing project information across its entire lifecycle, enabling facility condition monitoring through real-time data synchronization [7]. Numerous efforts are underway to enhance construction and maintenance efficiency through this integration [8,9], with BIM improving data interoperability and optimizing maintenance

processes for architectural asset performance [7]. Furthermore, BIM is evolving into a technology that bridges the physical and digital realms by merging real-time data with static models [10]. Recent studies suggest that integrating BIM with digital twins can significantly enhance Life Cycle Sustainability Assessments (LCSAs) and maintenance efficiency [11,12]. Against this backdrop, there is a pressing need for innovative approaches to overcome the limitations of traditional static maintenance methods and enable real-time monitoring and predictive maintenance in urban infrastructure.

1.2. Motivation and Objectives

Noise Barrier Tunnels (NBTs) are essential urban infrastructures that mitigate traffic noise, requiring advanced technologies for effective long-term maintenance and performance [13,14]. However, the current methods, which rely on periodic visual inspections and passive maintenance, fail to capture real-time conditions, reducing efficiency and compromising safety [7]. The 2022 fire at the North Uiwang Interchange NBT on South Korea's Second Gyeongin Expressway, which caused five deaths and over 40 injuries, highlights the urgent need for systematic maintenance and real-time monitoring [15].

Despite increasing interest, the integrated use of BIM and digital twins in infrastructure like NBTs remains limited. While prior studies have explored BIM and IoT integration in construction and maintenance workflows, most remain theoretical with few real-world applications. For NBTs, which require continuous oversight, research on digital twin applications is particularly needed [13]. Most existing work focuses on conceptual frameworks for data integration using BIM and IoT, yet practical validation in infrastructure contexts remains inadequate [7]. In contrast, combining BIM and digital twins offers clear advantages, such as intuitive 3D-based maintenance, real-time condition monitoring via IoT, and predictive maintenance support [16]. However, applying these integrated technologies to NBTs is still in its nascent stages [17]. The utilization of real-time sensor data within a BIM-based digital twin facilitates the comparison of actual structural or environmental conditions with predefined performance thresholds in the model. In the event of the detection of anomalies, such as abnormal vibration, pitch variation, or irregular illuminance, the system provides immediate feedback to enable preventive action.

Therefore, this study proposes the development of a BIM-based digital twin monitoring system for solar-powered NBTs in South Korea. The system aims to overcome the limitations of traditional periodic inspections, namely their inability to capture dynamic changes and to improve maintenance efficiency by supporting early anomaly detection through data analysis.

1.3. Research Scope and Methodology

To surmount the limitations of conventional periodic inspections and enhance the efficiency of maintenance, this study developed and implemented a BIM-based digital twin monitoring system for solar-powered NBTs in South Korea. The research commenced with a review of extant studies on digital twins and BIM to define the project's objectives and requirements. A 3D BIM model of the tunnel and a design automation tool were created using Autodesk Revit and Dynamo. An IoT sensor network was then deployed to collect real-time data, including but not limited to vibration, tilt, illuminance, air quality, and water detection, from both the interior and exterior of the tunnel.

The collected data were stored in CSV and JSON formats and synchronized with the BIM model to dynamically reflect the tunnel's condition within the digital twin environment. The sensor readings were interpreted in two distinct ways—firstly, as direct measurements, and secondly, as values compared against predefined baselines or thresholds according to the sensor type. For instance, initial tilt angles were used to detect

abnormal pitch changes, and vibration intensity was assessed relative to safe operating limits. While real-time alerts are delivered through the dashboard based on detected deviations, the BIM model provides a spatial reference to contextualize sensor locations and conditions. Therefore, anomalies are interpreted through the data stream itself, rather than being directly visualized within the 3D model.

The overall methodology is outlined in Figure 1. It began with a literature review, followed by the development of the BIM model and the sensor-integrated digital twin system. After deploying the system on a solar-powered NBT, data were collected and partially visualized in real time via a web-based interface using the Autodesk Forge API. While basic alerts based on device-reported flags were displayed on the dashboard, a more detailed evaluation of anomaly patterns, such as irregular time intervals or abnormal sensor readings, was performed offline using raw data analysis. This two-tiered approach allowed for verification of data integrity and sensor reliability under field conditions. The final phase of the methodology evaluated the system's integration performance and its ability to support real-time monitoring and maintenance decision-making, providing insight for future enhancements.

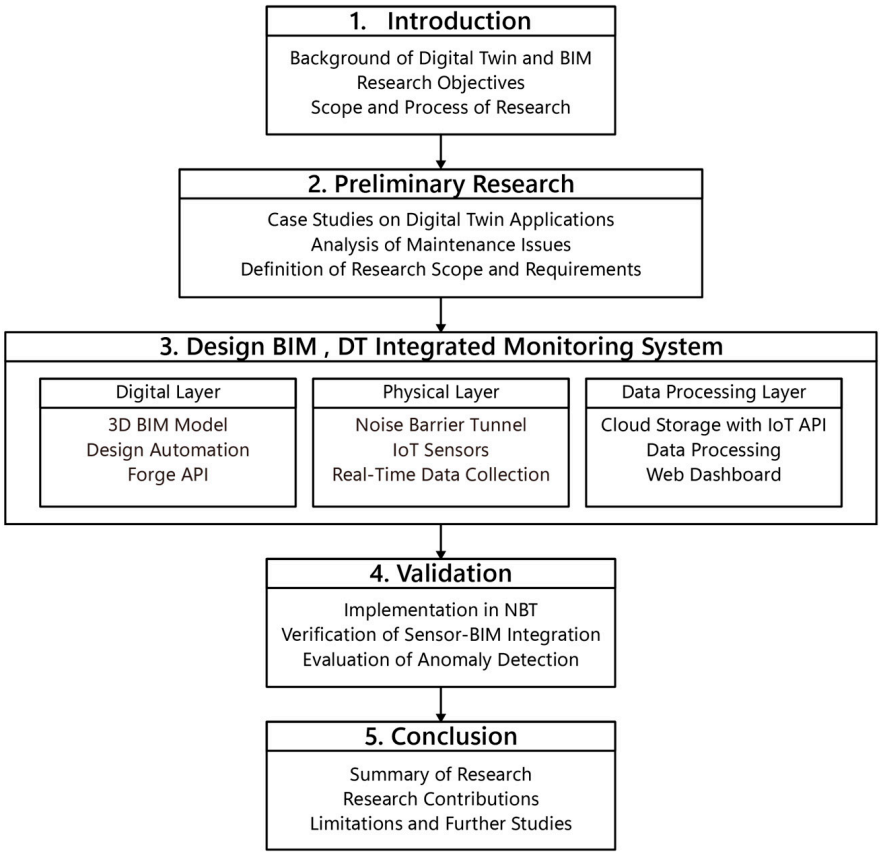


Figure 1. Research methodology and process.

2. Literature Review

2.1. Integration of BIM and Digital Twin Concepts

A digital twin is an advanced technology that integrates real-time data with a virtual representation of a physical object or system to enable continuous monitoring and predictive analysis. Originally introduced by Grieves in 2002, and further developed by NASA in 2010, the digital twin concept has evolved into a dynamic system for prediction and operational optimization [3,4,18]. Initially defined as a triad consisting of the physical entity, its digital counterpart, and the data interface, the digital twin has matured through the adoption of

IoT, Big Data, and AI technologies [19,20]. These advances have extended its application beyond manufacturing to infrastructure domains, such as tunnels, NBTs, and underground facilities, where it supports accurate structural assessment and proactive maintenance strategies [21,22].

Building Information Modeling (BIM) is a digital framework that captures the three-dimensional geometry, material properties, and construction sequence of buildings and road infrastructure, enabling lifecycle data management. Although BIM is inherently static, its integration with digital twins transforms it into a dynamic environment capable of reflecting real-time conditions via sensor and operational data [13,23]. Recent studies have shown that BIM provides the foundational layer for digital twin environments, where the integration of IoT-based sensing and AI-driven predictive algorithms significantly improves operational monitoring and safety management, particularly in tunnel and NBT applications [24–27]. Continuous condition updates in this integrated environment improve maintenance decision-making and infrastructure resilience.

In addition, Kritzinger et al. (2018) categorized digital twins in manufacturing and affirmed their potential in predicting and maintaining complex systems [28]. However, their focus on manufacturing highlights the need for further exploration in the context of construction infrastructure. Kim and Kim (2020) applied digital twins to predict the lifetime of NBT components, demonstrating how they address the limitations of periodic inspections and validate the integration of BIM with IoT-based monitoring [13]. Similarly, Xu et al. (2021) demonstrated that real-time prediction and maintenance models based on digital twins can enhance the management of complex systems, providing valuable insights for the system design in this study [20].

Interdisciplinary advances are essential for the development of BIM-DT frameworks, especially for NBT monitoring. Alshammari et al. (2021) highlighted the importance of cybersecurity, such as encryption and authentication, to protect real-time sensor data, which is critical for NBT reliability [29]. Lopez and Akundi (2022) proposed model-based systems engineering (MBSE) to integrate IoT, analytics, and BIM within a unified architecture to optimize system performance [30]. Soman et al. (2025) highlighted the role of human–machine interfaces in improving operator decision-making through actionable insights [31]. Lampropoulos et al. (2024) emphasized the contribution of digital twins to NBT resilience and safety through predictive maintenance [32], while Noroozinejad et al. (2024) extended the scope by incorporating digital twins into immersive metaverse environments to support enhanced design review, training, and lifecycle management [33].

In summary, the integration of BIM and digital twins enables dynamic infrastructure representation that goes beyond the static limitations of BIM alone. Previous studies and interdisciplinary developments confirm the value of this approach in improving maintenance efficiency and safety. Building on these findings, the present study demonstrates the feasibility and practical utility of this approach.

2.2. Limitations of Traditional Maintenance Approaches

Traditional infrastructure maintenance relies primarily on scheduled visual inspections and manual assessments to evaluate asset conditions and determine necessary interventions. However, this approach has several limitations. First, fixed inspection cycles make it difficult to detect sudden structural changes or rapidly developing damage in a timely manner [20,22]. Second, inspections rely heavily on human judgment, making the process labor-intensive, error-prone, and inconsistent, which in turn increases maintenance costs and reduces emergency response [23,34]. Third, critical infrastructure, such as tunnels, underground spaces, and bridges, are particularly vulnerable to fatigue and degradation caused by environmental and traffic variations that may occur between inspections but

remain unmonitored [21]. Fourth, the lack of real-time data and digital integration results in poor coordination and slow decision-making [35].

Previous studies have highlighted these shortcomings. For example, Zhao et al. (2022) reported that traditional periodic inspections are insufficient for tunnel safety because they cannot capture dynamic structural changes [25]. Wang et al. (2024) also found that traditional maintenance of large underground facilities fails to integrate dispersed information in real time, leading to inefficient and reactive management [26]. These limitations are particularly critical in infrastructure environments, such as tunnels, NBTs, and underground spaces, where internal and external conditions are highly variable and complex. Recent studies have addressed these challenges by introducing dynamic maintenance frameworks that incorporate digital twin and IoT technologies [21,27]. Such approaches are expected to play a key role in overcoming the limitations of traditional methods, especially for NBTs, which require customized maintenance strategies due to their unique structural and operational characteristics [7]. The 2022 fire at the North Uiwang Interchange NBT on South Korea's Second Gyeongin Expressway, which resulted in five deaths and over 40 injuries, highlights the urgent need for real-time monitoring and systematic maintenance [15].

The BIM-based digital twin monitoring system developed in this study responds to these needs by combining static BIM design data with real-time sensor inputs to continuously update the condition of solar-powered NBTs. This integration enables rapid anomaly detection and response, while reducing reliance on manual labor and minimizing data inconsistencies. Ultimately, it supports a more systematic and proactive maintenance and safety management framework.

2.3. Review of Prior Research

Prior studies have shown that the integration of digital twins with BIM goes beyond the replication of 3D models, enabling real-time monitoring and dynamic maintenance through sensor and IoT technologies. Grieves and Vickers laid the theoretical foundation for digital twins by introducing the concept of data-driven connectivity between physical and virtual systems [18]. Building on this, Tao et al. (2018) explored the evolution of digital twins with IoT and AI integration, highlighting their potential for real-time condition prediction and proactive maintenance [19].

In the infrastructure domain, Zhao et al. (2022) highlighted the role of sensor data in detecting sudden structural changes during tunnel safety assessments [25], while Wang et al. (2024) demonstrated that integrating distributed data streams improves consistency and responsiveness in managing large underground facilities [26]. Focusing on NBTs, Kim and Kim (2020) proposed fatigue-based lifetime prediction models for tunnel components, offering insights into structural performance but limiting their scope to component-level analysis without comprehensive real-time monitoring [13]. Mohammadi et al. (2023) demonstrated the integration of BIM with real-time vibration monitoring for bridge asset management, reducing inspection frequency and enhancing operational efficiency [36]. Similarly, Kaewunruen et al. (2020) applied digital twins to subway stations, achieving cost savings and sustainability improvements through real-time audits [37]. Zhong et al. (2023) provided an overview of predictive maintenance using digital twins, emphasizing algorithmic advancements for fault prediction and decision-making across industries, including infrastructure [38]. Wang et al. (2024) further reviewed the adoption of digital twins in construction projects, identifying opportunities for optimizing maintenance workflows and addressing implementation challenges [39]. While these studies offer robust frameworks and practical applications, empirical validation of fully integrated BIM–digital twin systems in real-world infrastructure settings remains limited. To address this gap, the present study

evaluates the effectiveness of a BIM-based digital twin monitoring system implemented in a solar-powered NBT to support real-time maintenance and safety management in the field.

Additionally, Kritzinger et al. (2018) proposed a classification framework for digital twins in manufacturing, highlighting their predictive maintenance potential, though its focus on manufacturing limits direct applicability to infrastructure [28]. Xu et al. (2021) introduced a digital twin-based model for performance optimization and predictive maintenance in aviation systems, demonstrating real-time applicability to complex infrastructure [20]. Yu et al. (2021) presented a digital twin framework for tunnel maintenance decision-making, achieving notable efficiency improvements through integrated real-time data and predictive analytics [40]. While these studies showcase the potential of real-time, IoT-based digital twin solutions, most remain confined to specific components or simulation environments, lacking comprehensive on-site validation. This study addressed these gaps by implementing a BIM-based digital twin monitoring system in an operational NBT context, tailored to site-specific conditions, thus laying the groundwork for practical field adoption and future integration of advanced analytics.

Table 1 summarizes key prior studies, which suggest that the integration of digital twins with BIM supports maintenance management by leveraging real-time sensor data and advanced analytics. While some research remains industry-specific—focused on manufacturing or aviation—recent studies have begun addressing civil infrastructure, including bridges, subway stations, and construction projects. However, empirical validation in fully integrated, complex infrastructure contexts, such as tunnels and NBTs, remains limited. To address this gap, the present study implements a BIM-based digital twin monitoring system in an actual NBT setting, drawing on the findings and limitations identified in Table 1. This empirical application demonstrates the practical feasibility of BIM-DT integration for real-time maintenance and safety management in urban infrastructure, while laying the foundation for future expansion into predictive algorithms and advanced analytics.

Table 1. Summary of prior research.

Author	Research Focus	Application Areas and Strengths	Shortcomings	Implications for This Study
Kim and Kim (2020) [13]	Fatigue-based life prediction	NBT components; structural assessment	Limited to fatigue analysis	Basis for real-time NBT monitoring
Yu et al. (2021) [40]	Decision support framework	Tunnel O&M; data integration	Scalability unverified	Validates system integration
Wang, H. et al. (2024) [26]	DT for underground spaces	O&M management; real-life validation	Specific to underground spaces	Empirical O&M applicability
Mohammadi et al. (2023) [36]	BIM-DT for bridge management	Bridge O&M; reduced inspection frequency	Focused on bridges, not tunnels	Real-time monitoring validation
Kaewunruen et al. (2020) [37]	DT for subway sustainability	Subway stations; cost efficiency	Specific to subway infrastructure	Supports cost-effective maintenance
Zhong et al. (2023) [38]	Predictive maintenance with DT	Multi-industry; algorithmic advancements	General overview, not NBT-specific	Framework for predictive analytics
Wang, M. et al. (2024) [39]	DT in construction projects	Construction O&M; workflow optimization	Broad review, lacks specific cases	Supports practical DT adoption
Kritzinger et al. (2018) [28]	DT classification	Manufacturing; maturity assessment	Manufacturing-focused	Foundation for construction O&M
Tao et al. (2019) [19]	DT application review	Cyber-physical integration; analytics	Construction specificity lacking	Identifies core DT technologies
Xu et al. (2021) [20]	DT optimization	Aviation; real-time feedback	Limited to aviation	Real-time optimization potential

3. Design of a Noise Barrier Tunnel Maintenance Monitoring System

Traditional maintenance practices for NBTs rely primarily on periodic visual inspections, which are inadequate for detecting real-time structural or environmental changes. These limitations can lead to delayed responses, increased costs, and safety risks. While BIM has been widely adopted during design and construction, its integration with real-time data during operational phases remains limited. To address this gap, this study introduced a digital twin-based monitoring system that integrates BIM and IoT sensor data to improve real-time condition monitoring and predictive maintenance in NBTs. This section presents the system architecture, its component layers, and the implementation strategy adopted to support practical applicability in field conditions.

3.1. System Overview and Objectives

This study developed a BIM-based digital twin monitoring system and demonstrated its potential to improve maintenance efficiency and safety in NBTs. By integrating BIM, DT technologies, and IoT sensors, the system addresses the limitations of conventional periodic inspections and enables real-time condition tracking and predictive maintenance. The BIM model provides static design data, such as geometric configuration and material specifications, while the DT integrates this with real-time environmental and structural data (e.g., temperature, vibration) collected by IoT sensors. This integration enables dynamic monitoring of tunnel conditions in a virtual environment, facilitating early anomaly detection and data-driven decision-making.

The system architecture consists of three interrelated layers, as shown in Figure 2. The digital layer establishes a maintenance-oriented environment using a 3D BIM model developed in Autodesk Revit at LOD 400, supported by design automation workflows. The physical layer includes the actual NBT and a sensor network that measures key parameters, including temperature, humidity, vibration, and light levels. The data-processing layer transmits and stores sensor data via a commercial IoT API, pre-processes them, and synchronizes them with the BIM model in real time. The data are transferred using WebSocket protocols, and visualization is enabled through the Autodesk Forge API and a web-based dashboard. This layered framework supports continuous monitoring and system responsiveness, positioning BIM-DT integration as a practical solution for predictive maintenance in urban infrastructure.

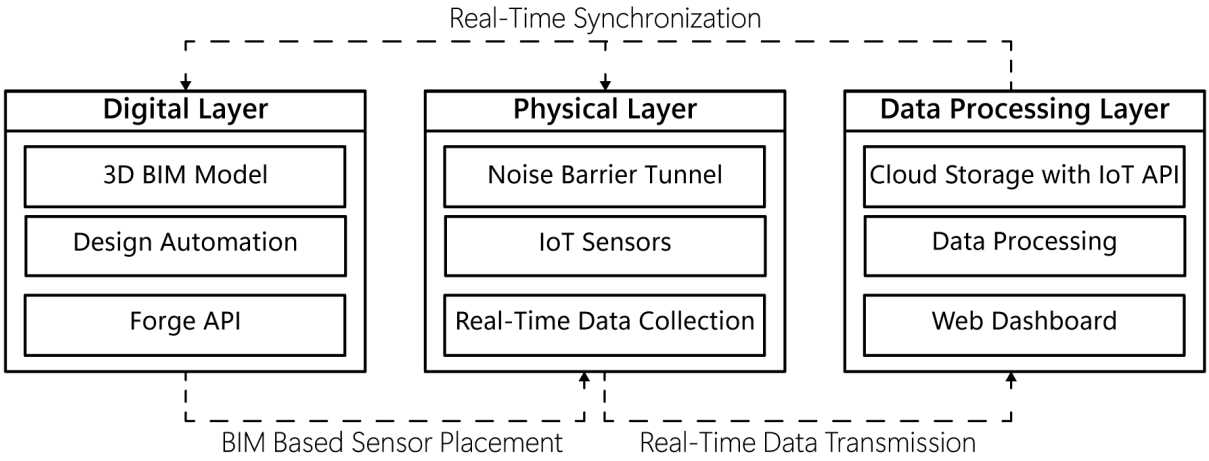


Figure 2. Main layer composition of the NBT maintenance monitoring system.

Technically, the system uses Autodesk Revit and Dynamo to develop a detailed BIM model at LOD 400, while design automation increases model flexibility and responsiveness. Real-time data transfer is enabled through WebSockets, and synchronization with the BIM

model is maintained through the Autodesk Forge API. This infrastructure supports a web-based dashboard that provides real-time visualization, anomaly detection, and decision support for maintenance operations.

This study helps overcome the limitations of traditional inspection-based maintenance through three key contributions: enhanced real-time monitoring, predictive maintenance capabilities, and improved maintenance efficiency and safety. By combining BIM and IoT within a unified platform, the system provides a structured framework for real-time, data-driven tunnel management. It drives the shift from reactive to predictive maintenance practices, providing practical value to infrastructure operators.

3.2. Digital Environment and BIM Model Development

Traditional BIM models have prioritized static data management during the design and construction phases, limiting their ability to reflect dynamic changes in infrastructure, such as NBTs. This lack of real-time integration hinders the timely detection of structural or environmental changes, reducing maintenance effectiveness and increasing operational risk. To overcome these limitations, this study established a digital environment that combines BIM with digital twin technology, enabling real-time synchronization, continuous condition monitoring, and predictive maintenance for NBTs.

A BIM model at LOD 400 was developed using Autodesk Revit, supported by a set of parametric modeling scripts created in Dynamo. As shown in Figure 3, the modeling process begins with the import of roadway geometry (Road.dwg) and predefined Revit family components. The scripting pipeline automatically generates the tunnel floor, structural members, panel systems, connectors, and rafter beam cover assemblies, systematically constructing the entire geometry with precision and consistency. The resulting model is saved in RVT format for integration and visualization and converted into SVF format for web-based interaction. The LOD 400 model includes detailed attributes, such as photovoltaic module dimensions, bolted joints, and structural framing, to support sensor placement and long-term asset management.

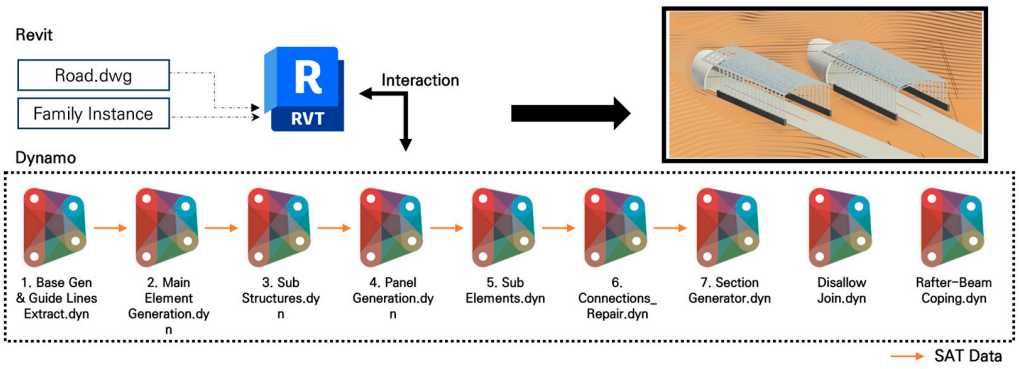


Figure 3. BIM model with LOD 400 level created using Autodesk Revit and Dynamo.

Real-time sensor data are collected from vibration sensors (mm/s^2), tilt detectors (degrees), air quality monitors (PM1, PM2.5, PM10 in $\mu g/m^3$), light sensors (lux), and water detection devices (Boolean). These values are transmitted to the cloud via a commercial IoT aggregation API, refined and validated through a sensor API, and synchronized with the BIM model using the Autodesk Forge API and WebSockets. This real-time streaming supports the visualization of current conditions and alerts on a web-based dashboard. As illustrated in Figure 4, the system architecture comprises three key stages. First, sensor data are acquired in real time from the physical tunnel environment. Second, the data are transmitted to a central repository via authenticated WebSocket communication, undergoing format conversion for compatibility. Third, the monitoring client integrates the LOD

400 BIM model and sensor data via the Forge API, rendering the model in SVF format for interactive visualization. This integration enables anomaly detection, facilitates intuitive monitoring, and supports data-driven maintenance decisions.

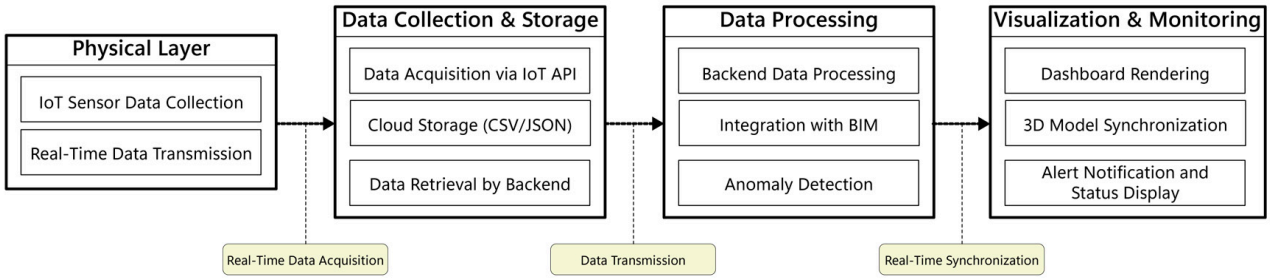


Figure 4. Process of transmitting and processing the acquired data.

This approach moves beyond traditional periodic inspections by enabling real-time responsiveness, reducing manual intervention, and enhancing safety and operational efficiency for urban infrastructure maintenance. Although the project did not follow a rigid linear development model, initial sensor testing and connectivity verification were conducted in a lab-scale prototype environment. This informal preliminary phase helped validate data acquisition, synchronization, and alert thresholds before full-scale deployment in the field. Future implementations may benefit from a more formalized testing framework and a clearly defined development timeline.

3.3. Physical Environment Setup and IoT Sensor Network

To establish a reliable real-time monitoring system for the NBT, this study implemented an integrated IoT sensor network within the physical tunnel environment. This network serves as the primary interface between the tunnel's real-world conditions and the digital twin system, enabling continuous monitoring and timely anomaly detection. The sensors collect structural and environmental data, which are then synchronized with the BIM model to reflect the dynamic state of the tunnel.

Five types of sensors were selected based on their relevance to tunnel safety, operational efficiency, and maintenance requirements: vibration sensors, tilt sensors, light sensors, air quality sensors, and water detection sensors. Each type of sensor captures a specific variable and transmits it in a standardized digital format suitable for real-time analysis and visualization. Table 2 summarizes the measured variables, data formats, and maintenance-related functions of each sensor.

Table 2. Overview of sensors.

Sensor Type	Measured Variable	Unit	Sampled Input Format	Maintenance Purpose in NBT	Twin Application
Vibration	Acceleration/intensity	mm/s^2, m/s^2	Float (e.g., 0.62)	Detects abnormal stress/crack	Structural health model
Tilt	Pitch angle (θ)	Degrees (°)	Float (e.g., 1.24°)	Detects foundation or frameshift	Deformation visualization
Light (Illuminance)	Illuminance	Lux	Integer (e.g., 1540 lux)	Checks visibility and solar panel output	Solar efficiency mapping
Air Quality	PM1, PM2.5, PM10	$\mu g/m^3$	Float (e.g., 42.1)	Assesses tunnel air safety	Environmental safety index
Water Detection	Flood presence (binary)	Boolean	True/false	Detects road flooding and blockage	Accessibility alert

To provide a clearer understanding of the sensor deployment, Figure 5 shows two views (frontal and isometric) of a single NBT. In the front view, vibration meters and

light meters are installed at the top truss level (Z: +11,072) to monitor structural vibration and light exposure from external sources. Tilt sensors, temperature sensors, air quality meters, and water detection sensors are distributed along vertical positions from Z: +5035 to Z: +1095 on the steel columns and floor to comprehensively monitor structural tilt, environmental conditions, and potential flooding at the tunnel floor. This placement strategy ensures effective coverage of critical zones, including the roof structure, access level ventilation, and flood-prone areas. All sensors are configured to collect data at 10-min intervals, balancing energy efficiency, transmission reliability, and maintenance responsiveness. The collected data are transmitted via WebSocket protocol to a cloud server, enabling near real-time synchronization with the digital twin platform. This supports dynamic visualization, automated anomaly detection, and maintenance decision-making through a unified monitoring interface. The same sensor layout and vertical positioning strategy is mirrored on the opposite road, where another NBT structure is installed. This ensures consistent monitoring in both tunnel directions.

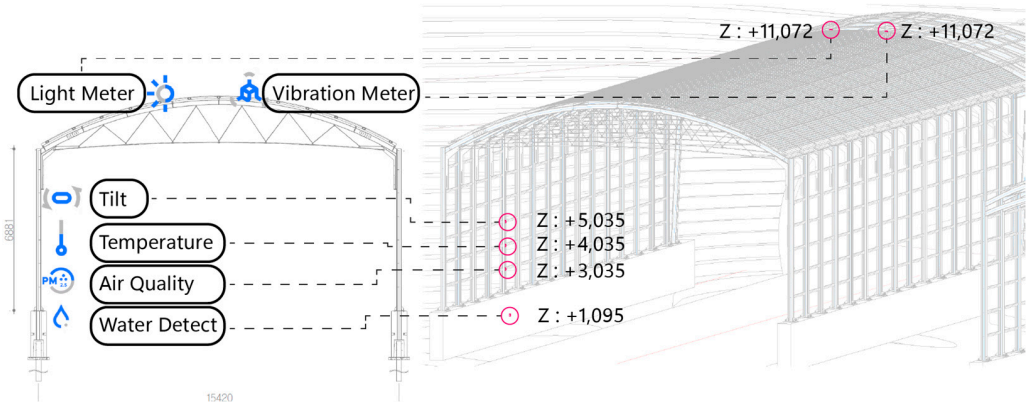


Figure 5. Overview of the installation of sensors in NBT (Tunnel 1).

The sensor network is designed to transmit collected data to a centralized cloud repository using a commercial IoT aggregation API. Its low-power architecture and robust communication protocols ensure long-term autonomous operation. Sensor readings are processed and delivered to the digital twin via a structured WebSocket stream, enabling near real-time updates on conditions such as vibration intensity, pitch angle variations, and the presence of surface water. Unlike traditional inspections that rely on infrequent site visits and may miss transient anomalies, this configuration supports continuous tracking and rapid anomaly detection based on predefined thresholds. This real-time responsiveness helps reduce maintenance costs and improve structural safety.

Figure 6 illustrates the data pipeline from sensor acquisition to cloud transmission and synchronization with the BIM-based monitoring interface, including integration via the Autodesk Forge API. Although the system does not employ equation-based numerical modeling, each sensor input is evaluated through logic-based rules implemented via API integration and condition monitoring functions. These rules use threshold comparisons and hardware status flags (e.g., Sensor State = 2) to determine real-time alert levels and system responses. Although the system does not use explicit numerical equations for state prediction, each sensor reading is interpreted through threshold-based logic, anomaly rules, and comparison to baseline values to represent behavior in the digital twin environment.

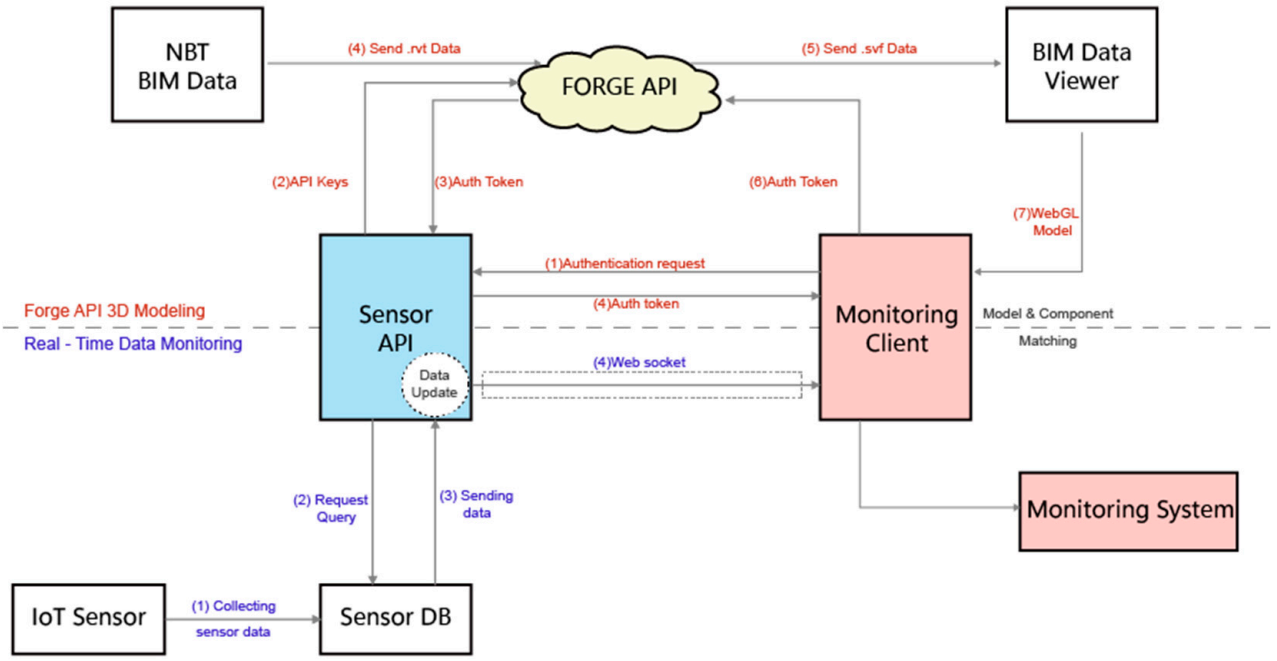


Figure 6. Composition and data flow of the sensor network.

Each sensor measures a specific physical variable, but several measurements are functionally related. For example, the light levels indirectly reflect solar irradiance, which is tied to photovoltaic performance. Similarly, the tilt and vibration sensors complement each other in detecting structural instability; tilt sensors are sensitive to gradual positional shifts, while vibration sensors capture dynamic impacts. By analyzing these inputs together, the system enhances the contextual understanding of the tunnel's condition within the digital twin.

3.4. Real-Time Monitoring System Design

The real-time monitoring system developed in this study enables continuous tracking of the physical condition of the NBT and supports timely maintenance decisions. As shown in Figure 6, sensor data are transmitted from the tunnel site to a central cloud repository, where they are processed and synchronized with the BIM model. This end-to-end data pipeline—from acquisition to visualization—is the backbone of the monitoring framework discussed in this section. Sensor readings are transmitted via a commercial IoT API and delivered to the client in real time using WebSocket communication. These data streams are parsed and structured in the cloud to support integration with the NBT's LOD 400 BIM model. The model is rendered using the Autodesk Forge API and converted into Simple Vector Format (SVF) for fast and interactive access via web browsers.

On the client side, a web-based dashboard provides intuitive monitoring of the system status, structural conditions, and environmental variables. The dashboard displays real-time sensor values, synchronized 3D views of the NBT, and visual alerts when anomalies occur. As shown in Figure 7, this interface integrates tabular data, live graphics, and interaction with the BIM model, allowing maintenance personnel to efficiently assess the condition of the tunnel. To ensure secure transmission and controlled access, all API communications are authenticated and encrypted using platform-level protocols, such as token-based access and TLS encryption. While this provides a basic level of security, more robust system-level features, including user role access control and data-level encryption, are not yet fully implemented. These enhancements have been identified as key priorities for future releases of the platform. By eliminating reliance on periodic inspections and

enabling continuous data-driven monitoring, the system provides a proactive framework for infrastructure maintenance and security management.

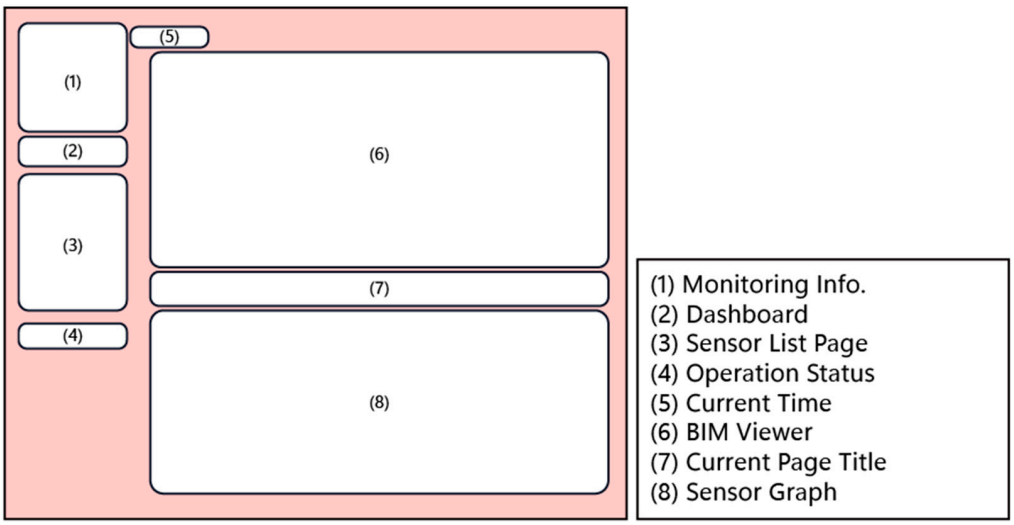


Figure 7. Dashboard configuration of the monitoring system.

4. Demonstration and Performance Assessment

4.1. Demonstration Targets and Site Conditions

This study focused on a solar-powered NBT integrated with the Wansan Tunnel in Jeonju, Republic of Korea. Designed to mitigate traffic noise and enhance safety in urban transportation infrastructure, the NBT serves as the demonstration target, as illustrated in Figure 8. The tunnel measures approximately 7~8 m in height and 14~17 m in width. Its geometric structure and component layout formed the basis for developing a detailed BIM model. A range of IoT sensors, including water detection, motion detection, tilt, temperature, air quality, and vibration sensors, were deployed throughout the tunnel to monitor structural and environmental conditions in real time (see Figure 5). Each sensor is supported by an optimized power supply and communication module, transmitting data to a centralized cloud repository via a commercial IoT data aggregation API.

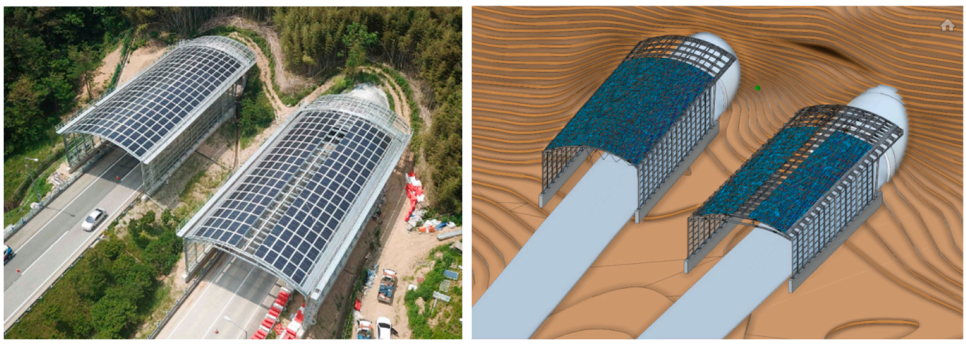


Figure 8. BIM-based digital model and physical implementation of the NBT on site. (Location: N 35°48′46.44″, E 127°11′29.4″).

The BIM model, developed at LOD 400 using Autodesk Revit, contains detailed geometric and operational data. It is synchronized with the digital twin environment using the Autodesk Forge API. This integration provides a robust foundation for real-time structural monitoring, anomaly detection, and maintenance support. The site's physical conditions and sensor deployment details serve as critical baseline data for system validation and

performance evaluation, demonstrating the practical applicability of BIM–digital twin integration in urban infrastructure management.

4.2. Prototype Integration and Functional Validation

Figure 9 illustrates the comprehensive process from the design and construction of the NBT to the implementation of the digital twin monitoring system. The top section shows the digital development pipeline, from element modeling and design automation, using Autodesk Revit and Dynamo, to BIM file conversion, using the Forge platform. The lower section depicts the physical implementation, showing the field installation of sensors and the real-time dashboard interface. This illustration shows the system's integration path from virtual modeling to field deployment and web-based visualization.

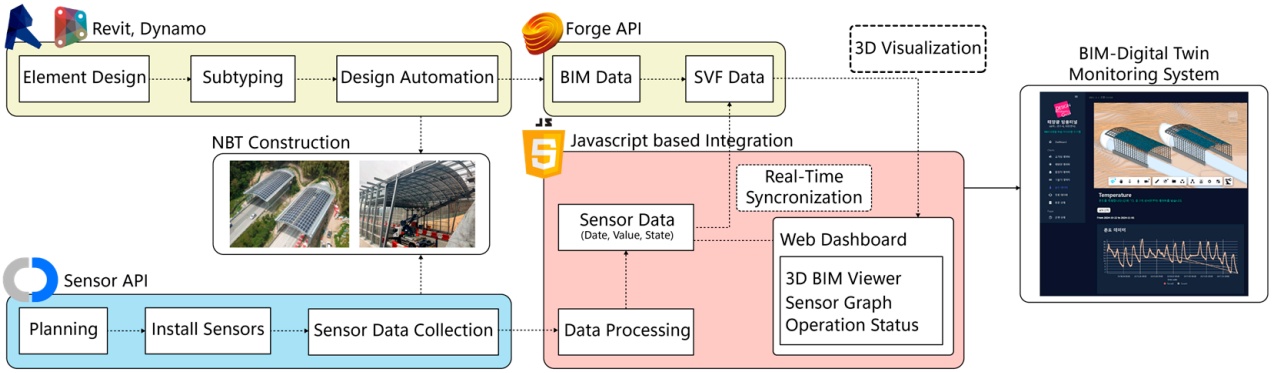


Figure 9. BIM and digital twin integrated monitoring system configuration process.

To ensure real-time monitoring, an array of IoT sensors was deployed inside and outside the NBT structure, including vibration, tilt, light, air quality, and water detection units. Figure 10a,b presents the hardware configuration of these sensors prior to deployment, with modular components prepared for various structural and environmental monitoring needs. These sensors serve as primary input nodes to the digital twin platform, providing the data backbone for anomaly detection and condition assessment. Figure 10c–e details the on-site sensor installation process, capturing how the sensors were mounted on the tunnel's interior steel columns and exterior surfaces. The images show practical considerations, such as exposure to sunlight for light sensors and ground-level placement for water detection. These visual records validate the system's field readiness and the integration of sensor networks with the tunnel's architectural elements.

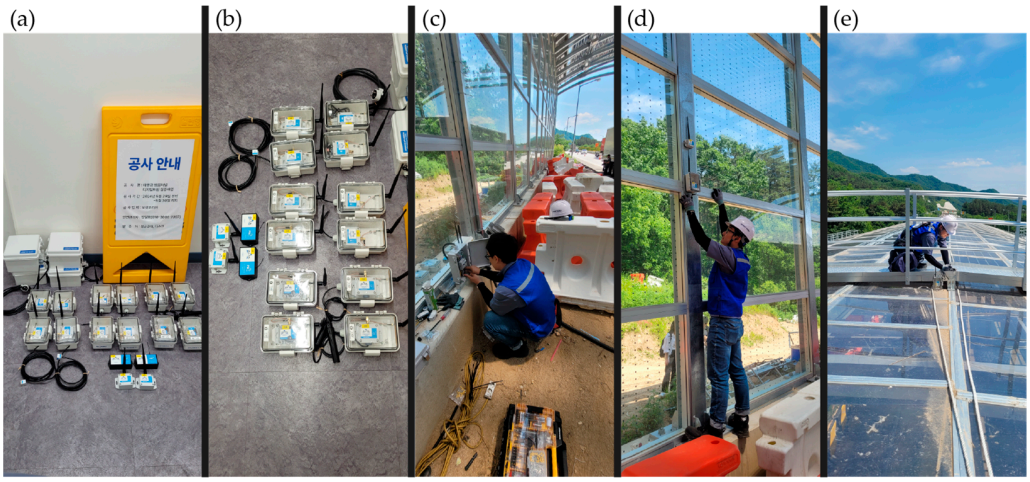


Figure 10. (**a**,**b**) Sensors prepared before installation. (**c**–**e**) Field installation of sensors in the NBT.

Sensor data are transmitted to a centralized cloud repository via a commercial IoT data aggregation API and synchronized with the BIM model via WebSocket communication, enabling real-time visualization and alerts. Initial post-installation testing confirmed stable sensor communication and real-time responsiveness, with no observable latency or data loss, supporting the system's potential for proactive maintenance management.

4.3. Results and Performance Analysis

To assess the reliability and performance of the deployed monitoring system, this section analyzes the sensor data collected during the demonstration period. While the light sensors initially provided the most consistent data stream, additional analysis confirmed that other sensors—vibration, tilt, air quality, and water detection—also transmitted data successfully and were visualized through the digital twin dashboard (Figure 11). This confirmed that the platform infrastructure and integration logic were operational across the sensor types.

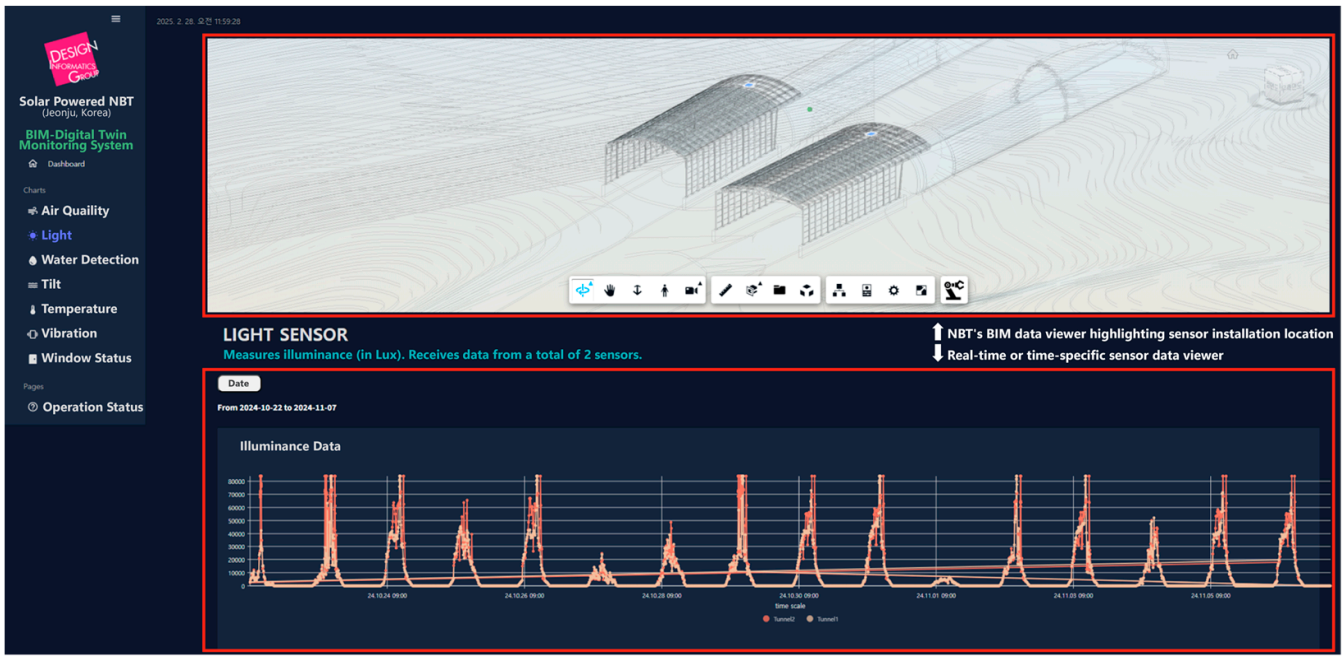


Figure 11. Visualization of the sensor data-based monitoring system.

Anomaly detection was performed using a complementary two-part approach. The first approach used system-reported status flags, such as Sensor State = 2, Alert Sent = True, and Battery = 0, which were monitored in real time via a dedicated status page on the dashboard. This mechanism enabled immediate operator notification in response to deviations from expected sensor ranges, such as sudden spikes or drops in illuminance, as shown in Figure 12. The second approach involved post hoc analysis of raw sensor data to identify indirect anomalies not flagged by device logic, including irregular sampling intervals, flatline trends, and unexpected environmental values. Although not embedded in the live monitoring platform, this analysis provided valuable diagnostic insights into data reliability and sensor behavior. Together, these two approaches strengthened the overall robustness of the system and provided direction for future enhancements.

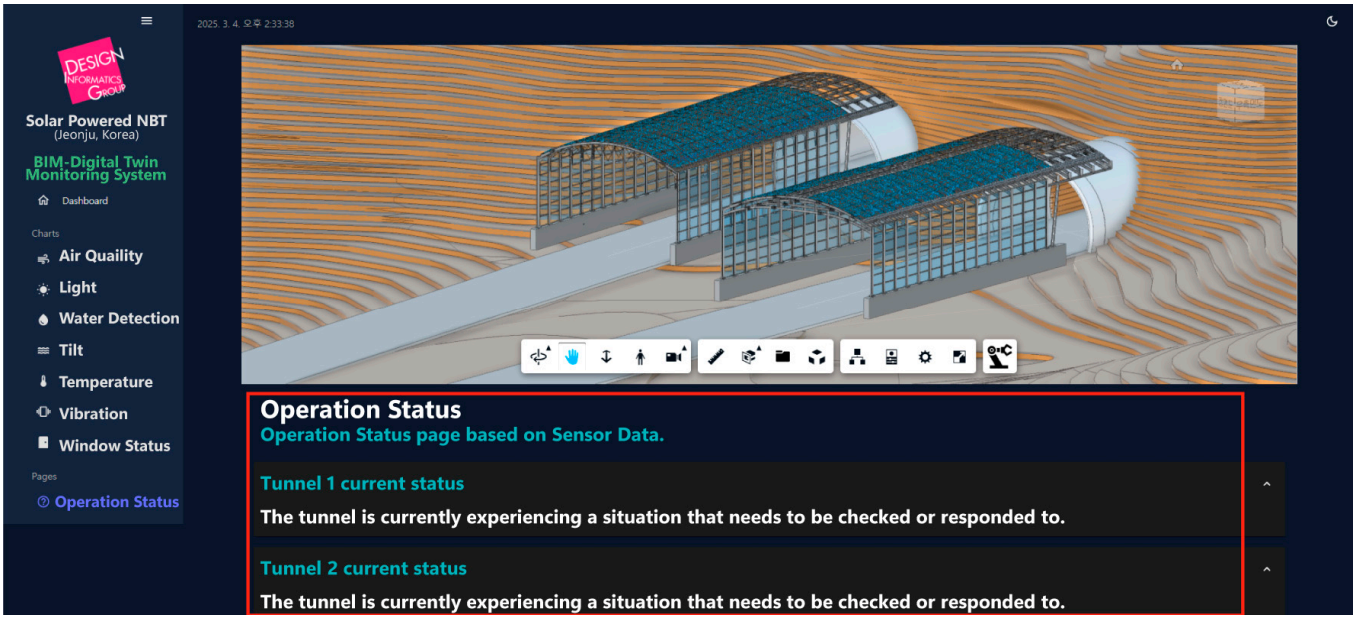


Figure 12. Sensor transmission data status: Data-based anomaly notification. (Translated from Korean to English).

Vibration sensors 01 and 02 exhibited stable performance throughout the demonstration period. No system-level alarms (e.g., Sensor State = 2) were triggered, and acceleration values remained within the expected limits. Data logging intervals were consistent, confirming both sensor integrity and communication reliability. Tilt sensors 01 and 02 recorded tilt values within a narrow band of $\pm 1^{\circ}$, well below the alert threshold of $\pm 3^{\circ}$. No abrupt changes were detected, indicating structural stability during the monitoring period. These patterns are illustrated in Figures 13 and 14, which show the vibration time series and pitch angle trends, respectively, with threshold values superimposed.

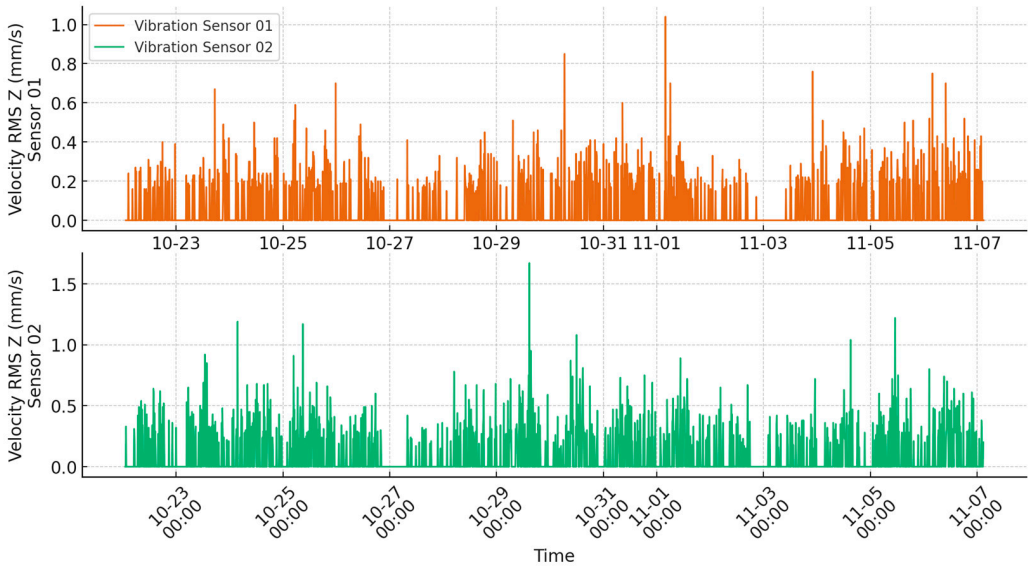


Figure 13. Vibration sensor: Velocity RMS Z.



Figure 14. Tilt sensor: Pitch angle variation.

Light sensors 01 and 02 achieved high data transmission rates of 92.1% and 97.0%, respectively, confirming regular synchronization with the cloud repository. However, offline analysis revealed anomalies such as 1-min recording intervals, repeated maximum illuminances (e.g., 83,865.6 lux), and zero lux values during daylight hours. These may be due to transient occlusions or processing delays at the sensor level. These anomalies are visualized in Figure 15, with shading indicating irregular intervals and abnormal values. Air Quality Sensor 01 transmitted 647 out of 933 expected records between 29 October and 5 November, resulting in 30.65% data loss. However, 99.07% of the data received was within the 10-min recording cycle, reflecting high temporal accuracy when active. Sensor 02 did not transmit any data. Figure 16 shows the PM1, PM2.5, and PM10 values over time, with shaded areas indicating gaps caused by irregular logging. Water Detection Sensors 01 and 02 recorded binary outputs with no flood detection events. Sensor 01 transmitted data consistently; sensor 02 did not transmit at all, indicating a persistent error. Although no actual flooding events occurred, the system's ability to record "false" baseline values verified its readiness to detect critical events.

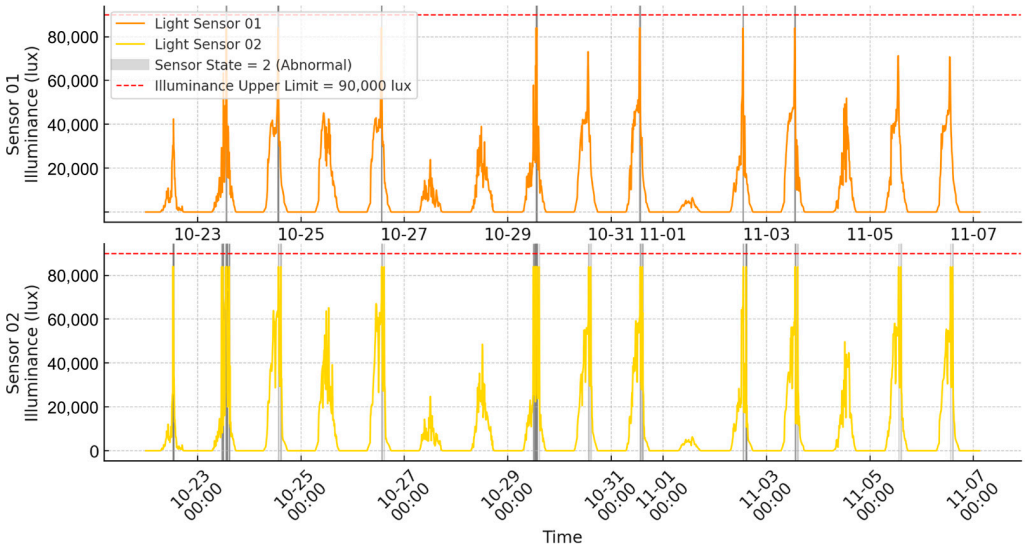


Figure 15. Light sensor: Illuminance with abnormality.

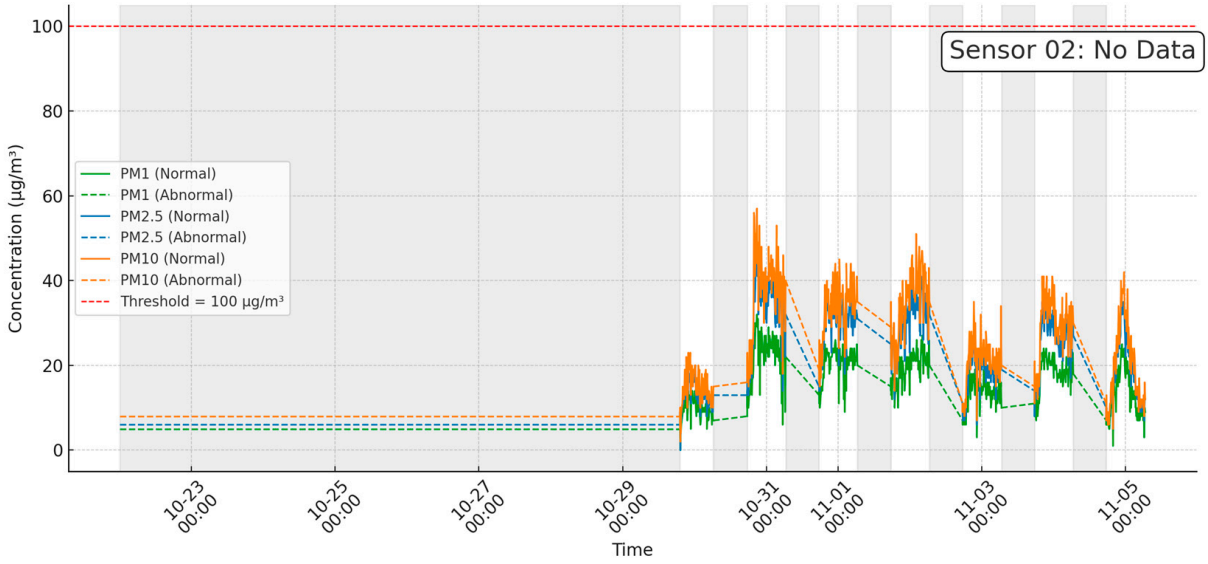


Figure 16. Air quality: PM trends with data gaps.

Anomaly assessment highlights the complementary value of hardware-level alerts and offline interpretation of raw data. In several cases, such as anomalous illuminance values or subtle tilt angle shifts, no system alert was triggered, yet post-analysis revealed clear anomalies. This underscores the importance of external validation to detect anomalies that may go unnoticed by device-level logic alone. Conversely, communication failures, such as those seen in Water Sensor 02, were immediately apparent from the lack of data and could be visualized directly on the monitoring platform. These findings support the integration of both alert-based and data-driven diagnostics for increased reliability.

Although the offline analysis in this study was performed retrospectively, the findings show clear potential for real-time application. Visual elements, such as threshold overlays, shaded data gaps, and highlighted anomalies, as illustrated in Figures 15 and 16, can be adapted for real-time anomaly detection and alert generation. Future iterations of the system may incorporate these techniques into the dashboard to support predictive maintenance workflows and enable proactive responses in tunnel operations scenarios.

4.4. Discussion and Future Enhancements

The demonstration confirmed that the proposed BIM-based digital twin monitoring system effectively integrates real-time sensor data with a high-resolution BIM model of the NBT site. The system employs a dual anomaly detection strategy, integrating direct alerts based on sensor-reported flags with offline analysis grounded in data-driven criteria. This integrated approach enhances the identification of hardware defects, environmental irregularities, and structural anomalies, providing significant advantages over conventional periodic inspections. Among the deployed sensors, light sensors exhibited the highest data coverage and demonstrated stable latency. The system's capacity to detect subtle anomalies beyond the thresholds of built-in sensor alerts is validated by additional insights from the vibration, tilt, air quality, and water detection sensors, including pitch deviations, illuminance anomalies, and irregular transmission intervals.

These outcomes align with prior research findings. For instance, Mohammadi et al. (2023) demonstrated that integrating BIM with real-time vibration monitoring reduced bridge inspection frequency and operational downtime [36]. Similarly, Kaewunruen et al. (2020) reported cost savings and efficiency gains in subway stations through BIM–digital twin platforms [38]. This study lends further support to the field applicability of digital twins for NBT maintenance.

Notwithstanding these strengths, this study has several limitations. While the dashboard successfully visualizes real-time sensor values and status flags (e.g., sensor state, battery status), more nuanced detections, such as gradual pitch drift or irregular illuminance cycles, require offline post-processing. This limitation restricts the system's responsiveness to emergent or low-severity conditions that do not trigger predefined thresholds. As Zhong et al. (2023) emphasized, effective predictive maintenance requires algorithmic intelligence beyond threshold-based alerts [38]. Additionally, while the system performed reliably in the NBT use case, its broader application to bridges, tunnels, and other asset types will necessitate calibration and tuning.

Future improvements should prioritize the incorporation of sophisticated analytics, including real-time pitch deviation detection, illumination pattern recognition, and inter-valve gap monitoring, directly into the monitoring platform. These enhancements would enable proactive maintenance through automated alerts based on contextual sensor trends, as suggested by Wang et al. (2024) [39]. User interface enhancements based on operator feedback, reduced data transmission latency, and enhanced cybersecurity protocols are also essential for long-term deployment. Given its demonstrated flexibility and effectiveness, the proposed framework is well-positioned for extension to a wider range of urban infrastructure applications. With these enhancements, the BIM–digital twin system can evolve into a core platform for data-driven, predictive asset management.

5. Conclusions

5.1. Research Summary and Key Contributions

This study developed and validated a real-time infrastructure monitoring system by integrating BIM and digital twin technologies at a solar-powered NBT site in Jeonju, Republic of Korea. The proposed system combined a high-resolution BIM model with real-time IoT sensor data, including vibration, inclination, illumination, air quality, and water presence. These data were synchronized through cloud-based architecture using the Autodesk Forge API and WebSocket communication protocols.

The system enabled continuous visualization and analysis of structural and environmental conditions and supported anomaly detection through a two-tiered approach: (1) direct alerts triggered by device-reported status flags, and (2) post-processed identification of anomalous trends, such as tilt angle drift, illuminance spikes, and irregular data intervals. These results demonstrate the system's ability to detect subtle irregularities, even when hardware-based thresholds were not exceeded. Real-time data transmission from the light sensors achieved rates of more than 90%, while offline analysis confirmed the stable performance of the tilt and vibration modules. Although air quality data coverage was partial, high interval fidelity underscored reliable timing and communication when active. This dual-insight strategy enhances situational awareness of emerging anomalies and complements traditional hardware-based monitoring.

These results are consistent with recent research highlighting the benefits of integrating sensor data into BIM-based systems. Mohammadi et al. (2023) demonstrated that integrating vibration monitoring into BIM models reduced inspection frequency and downtime in bridge infrastructure [36]. Similarly, Kaewunruen et al. (2020) demonstrated cost savings and efficiency gains in subway stations through digital twin applications [37], while Wang et al. (2024) emphasized the practical utility of digital twins in optimizing maintenance workflows in construction projects [39]. In addition, Zhong et al. (2023) highlighted the role of digital twin-based predictive maintenance in leveraging algorithmic intelligence to overcome the limitations of threshold-based approaches and provide a robust framework for real-time failure prediction and decision-making [38].

The key contributions of this study are as follows:

First, it implemented and validated a BIM–digital twin infrastructure monitoring system in a real-world context, integrating high-resolution BIM models with real-time environmental and structural sensor data.

Second, it demonstrated stable data collection, cloud-based synchronization, and integrated visualization via a web-based dashboard, confirming the feasibility of the system for real-time condition monitoring and maintenance support in complex infrastructure environments.

Third, sensor interoperability and data reliability were empirically evaluated under field conditions, confirming the robustness of the system architecture and its adaptability to infrastructure environments.

Taken together, these results highlight the pragmatic benefits of BIM–digital twin integration in enhancing situational awareness, enabling preventive maintenance, and improving operational efficiency in infrastructure management.

5.2. Study Limitations

The proposed system successfully demonstrated real-time integration of BIM and IoT sensor data in an operational NBT. However, several limitations emerged that constrain its full practical application. Most notably, intermittent power supply issues during the demonstration led to data loss and delayed sensor activation, particularly for the air quality and tilt modules. For instance, Air Quality Sensor 01 did not commence data collection until the first week of the demonstration and exhibited approximately 69.3% coverage during its active period. Despite the implementation of proper synchronization mechanisms, sporadic network instability also contributed to temporal discontinuities in the data stream.

Another key constraint lies in the limited real-time diagnostic capability of the current dashboard implementation. While the system does support immediate alerts based on sensor-reported flags (e.g., battery failure or sensor state anomalies), more advanced anomaly patterns (e.g., gradual pitch angle drift or irregular illuminance cycles) require offline post-analysis. This architectural separation, therefore, limits the system's capacity to provide immediate, data-driven maintenance suggestions. Furthermore, the demonstration was conducted on a single infrastructure type and site, which limits the generalizability of the findings. To ensure the effectiveness of the system in real-world scenarios, further validation is necessary to ascertain the compatibility of sensors with diverse infrastructure types, such as bridges or metro corridors, as well as to determine environment-specific thresholds and the system's long-term operational resilience.

Finally, the current system is deficient in the realm of real-time recovery and buffering mechanisms, which are imperative for the management of transient data outages resulting from power or network disruptions. Without such mechanisms, the reliability and precision of monitoring may be jeopardized under field conditions.

5.3. Direction for Future Research

To enhance the scalability and utility of the proposed BIM–digital twin system, future research should prioritize the integration of real-time analytics and predictive models. While this study applied threshold-based and interval analyses offline, the incorporation of such logic into the live platform will enable automated anomaly detection and early intervention. As Kaewunruen et al. (2021) observed, predictive maintenance necessitates dynamic analysis that extends beyond basic alerts. Secondly, the validation of the system across diverse infrastructure types, including bridges and tunnels, will assess its scalability and adaptability [37]. It is acknowledged that each context may necessitate customized thresholds and sensor configurations, and the broader implementation will contribute to the generalization of BIM–digital twin frameworks. Thirdly, enhancing the user experience and system resilience is imperative. Operator feedback has underscored the need

for more intuitive dashboards, customized alerts, and real-time visualization. Future systems should incorporate secure data transmission, buffering, and failover mechanisms to ensure continuity during power or network disruptions. Finally, the establishment of interoperable data-sharing frameworks could facilitate collaboration among cities, operators, and researchers. In accordance with Wang et al. (2024), the implementation of standardized platforms would expedite the adoption of digital twin technologies and facilitate the development of more comprehensive predictive maintenance strategies [39].

In summary, advancing analytics, expanding deployment, and improving usability and security are key to realizing the full potential of BIM–digital twin systems as robust infrastructure management platforms.

Author Contributions: S.-W.Y. conducted research planning and field demonstrations, wrote the manuscript, planned the initial monitoring system, and developed and executed the NBT BIM data creation script. Y.L. developed the API interpretation and monitoring system, and processed and visualized the collected sensor data. S.-A.K. supervised the research. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: Due to national research funding policies and infrastructure security regulations, the full BIM model and real-time monitoring dashboard cannot be publicly released. These models contain structural and proprietary information that was developed in collaboration with public infrastructure authorities and private companies in South Korea. However, anonymized sample sensor data collected during the demonstration period may be made available upon reasonable request to the author.

Conflicts of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this study.

References

1. Cho, Y. Criteria for Smart Cities, Smart City Indexes. *Sci. Technol. Innov. Policy* **2016**, *26*, 38–43. Available online: https://www.stepi.re.kr/site/stepiko/PeriodicReportList.do?cateCont=A0504&pageIndex=2&reIdx=0&searchYear=&searchCondition=1&searchKeyword=&searchSort=PUBLIC_DT#modal1 (accessed on 30 December 2024).
2. El-Agamy, R.F.; Sayed, H.A.; ALAkhatatneh, A.M.; Aljohani, M.; Elhosseini, M. Comprehensive Analysis of Digital Twins in Smart Cities: A 4200-Paper Bibliometric Study. *Artif. Intell. Rev.* **2024**, *57*, 154. [CrossRef]
3. Grieves, M.W. Product Lifecycle Management: The New Paradigm for Enterprises. *Int. J. Prod. Dev.* **2005**, *2*, 71–84. [CrossRef]
4. National Aeronautics and Space Administration (NASA). *Encyclopedia of Geography*; National Aeronautics and Space Administration (NASA): Washington, DC, USA, 2010. [CrossRef]
5. Dave, B.; Kubler, S.; Främling, K.; Koskela, L. Opportunities for Enhanced Lean Construction Management Using Internet of Things Standards. *Autom. Constr.* **2016**, *61*, 86–97. [CrossRef]
6. Dave, B.; Buda, A.; Nurminen, A.; Främling, K. A Framework for Integrating BIM and IoT through Open Standards. *Autom. Constr.* **2018**, *95*, 35–45. [CrossRef]
7. Nour El-Din, M.; Pereira, P.F.; Poças Martins, J.; Ramos, N.M.M. Digital Twins for Construction Assets Using BIM Standard Specifications. *Buildings* **2022**, *12*, 2155. [CrossRef]
8. Cheng, J.C.; Chen, W.; Chen, K.; Wang, Q. Data-Driven Predictive Maintenance Planning Framework for MEP Components Based on BIM and IoT Using Machine Learning Algorithms. *Autom. Constr.* **2020**, *112*, 103087. [CrossRef]
9. Teizer, J.; Wolf, M.; Golovina, O.; Perschewski, M.; Propach, M.; Neges, M.; König, M. Internet of Things (IoT) for Integrating Environmental and Localization Data in Building Information Modeling (BIM). In Proceedings of the International Symposium on Automation and Robotics in Construction, Taipei, Taiwan, 28 June–1 July 2017; Volume 34. [CrossRef]
10. UK BIM Alliance. *BIM and Digital Twins [Positioning Statement]*; UK BIM Alliance: London, UK, 2021.

11. Boje, C.; Guerriero, A.; Kubicki, S.; Rezgui, Y. A Framework Using BIM and Digital Twins in Facilitating LCSA for Buildings. *J. Build. Eng.* **2023**, *76*, 107232. [CrossRef]
12. Opoku, D.G.J.; Perera, S.; Osei-Kyei, R.; Rashidi, M. Digital Twin Application in the Construction Industry: A Literature Review. *J. Build. Eng.* **2021**, *40*, 102726. [CrossRef]
13. Kim, J.; Kim, S.-A. Lifespan Prediction Technique for Digital Twin-Based Noise Barrier Tunnels. *Sustainability* **2020**, *12*, 2940. [CrossRef]
14. Son, J. Case Study on the Planning Procedure of Traffic Soundproof Facilities and Alternative Decision Making. Master's Thesis, Bukyeong University, Busan, Republic of Korea, 25 February 2009. Available online: https://repository.pknu.ac.kr:8443/handle/2021.oak/10689 (accessed on 31 December 2024).
15. Kim, J.-M.; Son, S.-B.; Kim, H.-B.; Cho, J.-W. Apocalyptic Highway Fire Exposes Dangers of Plastic Tunnels. Available online: https://koreajoongangdaily.joins.com/2022/12/30/national/socialAffairs/korea-fire-gyeongin-expressway/20221230154034712.html (accessed on 28 December 2024).
16. Biagini, C.; Bongini, A. BIM and Data Integration: A Workflow for the Implementation of Digital Twins. In *Beyond Digital Representation. Digital Innovations in Architecture, Engineering and Construction*; Giordano, A., Russo, M., Spallone, R., Eds.; Springer: Cham, Switzerland, 2024; Volume 1, p. 822. [CrossRef]
17. Deren, L.; Wenbo, Y.; Zhenfeng, S. Smart City Based on Digital Twins. *Comput. Urban Sci.* **2021**, *1*, 4. [CrossRef]
18. Grieves, M.; Vickers, J. Digital twin: Mitigating unpredictable, undesirable emergent behavior in complex systems. In *Transdisciplinary Perspectives on Complex Systems. New Findings and Approaches*; Springer: Cham, Switzerland, 2017; pp. 85–113. [CrossRef]
19. Tao, F.; Zhang, H.; Liu, A.; Nee, A. Digital twin in industry: State-of-the-art. *IEEE Trans. Ind. Inform.* **2019**, *15*, 2405–2415. [CrossRef]
20. Xu, Z.; Ji, F.; Ding, S.; Zhao, Y.; Zhou, Y.; Zhang, Q.; Du, F. Digital Twin-Driven Optimization of Gas Exchange System of 2-Stroke Heavy Fuel Aircraft Engine. *J. Manuf. Syst.* **2021**, *58*, 132–145. [CrossRef]
21. Sychev, S.; Batako, A.; Nguyen, T.; Xavior, A.; Antosz, K.; Machado, J. An Investigation into Connection between BIM and Digital Twins Technologies. *IFAC-Pap.* **2024**, *58*, 234–239. [CrossRef]
22. Honghong, S.; Gang, Y.; Haijiang, L.; Tian, Z.; Annan, J. Digital twin enhanced BIM to shape full life cycle digital transformation for bridge engineering. *Autom. Constr.* **2023**, *147*, 104736. [CrossRef]
23. Boje, C.; Guerriero, A.; Kubicki, S.; Rezgui, Y. Towards a Semantic Construction Digital Twin: Directions for Future Research. *Autom. Constr.* **2020**, *114*, 103179. [CrossRef]
24. Pan, Y.; Zhang, L. A BIM-Data Mining Integrated Digital Twin Framework for Advanced Project Management. *Autom. Constr.* **2021**, *124*, 103564. [CrossRef]
25. Zhao, Y.; Cao, C.; Liu, Z. A Framework for Prefabricated Component Hoisting Management Systems Based on Digital Twin Technology. *Buildings* **2022**, *12*, 276. [CrossRef]
26. Wang, H.; Yu, C.; Zheng, J.; Jia, Y.; Liu, Z.; Yang, K. Digital-Twin-Based Operation and Maintenance Management Method for Large Underground Spaces. *Buildings* **2024**, *14*, 2577. [CrossRef]
27. Negri, E.; Fumagalli, L.; Macchi, M. A review of the roles of digital twin in CPS-based production systems. *Procedia Manuf.* **2017**, *11*, 939–948. [CrossRef]
28. Kritzinger, W.; Karner, M.; Traar, G.; Henjes, J.; Sihn, W. Digital Twin in manufacturing: A categorical literature review and classification. *IFAC-Pap.* **2018**, *51*, 1016–1022. [CrossRef]
29. Alshammari, K.; Beach, T.; Rezgui, Y. Cybersecurity for Digital Twins in the Built Environment: Current Research and Future Directions. *J. Inf. Technol. Constr.* **2021**, *26*, 159–173. [CrossRef]
30. Lopez, V.; Akundi, A. A Conceptual Model-based Systems Engineering (MBSE) Approach to Develop Digital Twins. In Proceedings of the 2022 IEEE International Systems Conference (SysCon), Montreal, QC, Canada, 25–28 April 2022; IEEE: New York, NY, USA, 2022; pp. 1–5. [CrossRef]
31. Soman, R.K.; Farghaly, K.; Mills, G.; Whyte, J. Digital Twin Construction with a Focus on Human Twin Interfaces. *Autom. Constr.* **2025**, *170*, 105924. [CrossRef]
32. Lampropoulos, G.; Larrucea, X.; Colomo-Palacios, R. Digital Twins in Critical Infrastructure. *Information* **2024**, *15*, 454. [CrossRef]
33. Noroozinejad Farsangi, E.; Shehata, A.O.; Rashidi, M.; Ghassempour, N.; Mirjalili, S. Transitioning from BIM to Digital Twin to Metaverse. *Front. Built Environ.* **2024**, *10*, 1486423. [CrossRef]
34. Liu, S.; Ren, S.; Jiang, H. Predictive Maintenance of Wind Turbines Based on Digital Twin Technology. *Energy Rep.* **2023**, *9*, 1344–1352. [CrossRef]
35. Lee, J.; Bagheri, B.; Kao, H. A Cyber-Physical Systems architecture for Industry 4.0-based manufacturing systems. *Manuf. Lett.* **2015**, *3*, 18–23. [CrossRef]

36. Mohammadi, M.; Rashidi, M.; Yu, Y.; Samali, B. Integration of TLS-derived bridge information modeling (BrIM) with a decision support system (DSS) for digital twinning and asset management of bridge infrastructures. *Comput. Ind.* **2023**, *147*, 103881. [CrossRef]
37. Kaewunruen, S.; Peng, S.; Phil-Ebosie, O. Digital twin aided sustainability and vulnerability audit for subway stations. *Sustainability* **2020**, *12*, 7873. [CrossRef]
38. Zhong, D.; Xia, Z.; Zhu, Y.; Duan, J. Overview of predictive maintenance based on digital twin technology. *Heliyon* **2023**, *9*, e14534. [CrossRef]
39. Wang, M.; Ashour, M.; Mahdiyar, A.; Sabri, S. Opportunities and threats of adopting digital twin in construction projects: A review. *Buildings* **2024**, *14*, 2349. [CrossRef]
40. Yu, G.; Wang, Y.; Mao, Z.; Hu, M.; Sugumaran, V.; Wang, Y.K. A Digital Twin-Based Decision Analysis Framework for Operation and Maintenance of Tunnels. *Tunn. Undergr. Space Technol.* **2021**, *116*, 104125. [CrossRef]

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